
LC Mathematics for Physicists 1B Lecture Notes

Year 1 Semester 2

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Table of Lectures

Lecture 1: Partial Differentiation I: Intro and Welcome	2
Lecture 2: Partial Differentiation II: The Total Differential	4
Lecture 3: Partial Differentiation III: Directional Derivatives and Gradients	7
Lecture 4: Partial Differentiation IV: Total Time Derivative and Multivariate Taylor Series	11
Lecture 5: Partial Differentiation V: 3D Stationary Points	14
Lecture 6: Partial Differentiation VI: Lagrange Multipliers	17
Lecture 7: Partial Differentiation VII: Statistical Mechanics	18
Lecture 8: Partial Differentiation VIII: Differentiation Under the Integral Sign	21
Lecture 12: End of Partial Differentiation & Start of ODEs	22
Lecture 13: ODEs II	25
Lecture 14	28
Lecture 15: Differential Equations III	30
Lecture 20: Methods of Integration	33

Wed 21 Jan 2026 11:00

Lecture 1 - Partial Differentiation I: Intro and Welcome

1 Course Welcome

1.1 Recommended books:

- Mathematical Techniques 4e, Jordan & Smith
- Engineering Mathematics 8e, Stroud
- Calculus (Schaum), 6e, Ayres & Mendelson
- Advanced Calculus (Schaum), 6e, Ayres & Mendelson

1.2 Assessment details:

- Maths 1A/1B form a single 20 credit module.
- 80% assessed by a 3 hour exam - Section 1 is 36% with 6 short questions and Section 2 is 64% with 4 long questions.
- 20% assessed by problem sheets.

1.3 Course structure:

1. Partial Differentiation
 - Definition, total differential, chain rule, gradient.
 - Taylor series, stationary points, Lagrange multipliers.
2. Differential Equations
 - Definition, 1st order separable, exact and homogenous.
 - Linear equations: general solution, 1st order and constant coefficients.
3. Integration
 - Definition as area under the curve, fundamental theorem of calculus.
 - Integration by: substitution, parts, partial fractions and tricks.
4. Multiple Integrals
 - Multiple and repeated integrals. Change of order of integration.
 - Change of variables and the Jacobian. Arc length. Solids of revolution.

2 Multivariate Functions

Lots of physics involves functions of more than one variable. A physical quantity defined at every point in space is called a field. We can have both scalar fields and vector fields.

For example, some scalar fields are:

- $V(x, y, z)$: Electrostatic potential. This is often easier to work with compared to the full electric (vector) field.

- $T(x, y, z)$: Temperature.
- $p(x, y, z)$: Pressure.

While some vector fields are:

- $\underline{E}(x, y, z)$: Electric Field.
- $\underline{B}(x, y, z)$: Magnetic Field.
- $\underline{v}(x, y, z)$: Velocity Field (i.e in fluid mechanics).

2.1 Partial Derivatives

Consider a function of two variables. The partial derivative of a function with respect to one variable is the rate of change of a function wrt that variable, while keeping other variables constant. Effectively, we carry out a derivative while treating the other variables as if they were constants.

Suppose we have a function $f(x, y)$. The definition of a partial derivative is:

$$\frac{\partial f}{\partial x}(x_0, y_0) = \lim_{h \rightarrow 0} \frac{f((x_0 + h), y_0) - f(x_0, y_0)}{h}$$

$$\frac{\partial f}{\partial y}(x_0, y_0) = \lim_{k \rightarrow 0} \frac{f(x_0, (y_0 + k)) - f(x_0, y_0)}{k}$$

Just like we denote $\frac{df}{dx}$ as the derivative of a function of a single variable, we denote $\frac{\partial f}{\partial x}$ as the partial derivative of a function of several variables.

Note that this is not delta f and delta y, i.e. not $\frac{\delta f}{\delta x}$

In theory, we'd explicitly notate:

$$\left(\frac{\partial f}{\partial x}\right)_y$$

With the subscript y explicitly stating that y is being kept constant. This is rarely, but sometimes, needed.

Consider $f(x, y, z) = x^2 \sin yz$. We have:

$$\frac{\partial f}{\partial x} = 2x \sin yz$$

$$\frac{\partial f}{\partial y} = x^2 z \cos yz$$

$$\frac{\partial f}{\partial z} = x^2 y \cos yz$$

2.2 Higher Orders

Higher derivatives are defined as they were previously, but they can now be mixed. For example, with $f(x, y) = x^2 \sin y$, we can write:

$$\frac{\partial f}{\partial x} = 2x \sin y \quad \frac{\partial f}{\partial y} = x^2 \cos y$$

$$\frac{\partial^2 f}{\partial x^2} = 2 \sin y$$

We can also have:

$$\frac{\partial^2 f}{\partial y \partial x} = \frac{\partial}{\partial y} \left(\frac{\partial f}{\partial x} \right) = 2x \cos y$$

Shorthand notation exists, i.e. $f_{xx} = \frac{\partial^2 f}{\partial x^2}$ or $f_{yx} = \frac{\partial^2 f}{\partial y \partial x}$

For most cases, but not all, mixed derivatives are often independent of the order of partial derivatives, so: $f_{xy} = f_{yx}$

Thu 22 Jan 2026 09:00

Lecture 2 - Partial Differentiation II: The Total Differential

1 The Total Differential

In order to generalise the chain rule, we need to define the total differential. Consider the change in a function of two variables, $f(x, y)$ when we move from some point (x, y) to some point $(x + dx, y + dy)$.

The partial derivative only tells us what happens when we change one variable, but we're changing two here. The total differential sums these two in order to get the full total change.

$$df = f(x + dx, y + dy) - f(x, y)$$

We have to look at the change in a single variable at a time, so we split it into two pieces where only x changes in the first, and only y changes in the second.

$$df = \underbrace{[f(x + dx, y + dy) - f(x, y + dy)]}_{\text{isolates change in } x} + \underbrace{[f(x, y + dy) - f(x, y)]}_{\text{isolates change in } y}$$

$$df = \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy$$

More generally for a function of $f(x_1, x_1, \dots, x_n)$:

$$df = \frac{\partial f}{\partial x_1} dx_1 + \frac{\partial f}{\partial x_2} dx_2 + \dots + \frac{\partial f}{\partial x_n} dx_n = \sum_{i=1}^n \frac{\partial f}{\partial x_i} dx_i$$

The total change in the function $f(x_1, x_1, \dots, x_n)$ is the sum of partial changes due to changing a single variable.

2 The Chain Rule

Recall that if $y = y(x)$, $x = x(t)$, then:

$$dy = \frac{dy}{dx} dx = \frac{dy}{dx} \frac{dx}{dt} dt \implies \frac{dy}{dt} = \frac{dy}{dx} \frac{dx}{dt}$$

Now, if $f = f(x, y)$ and $x = x(t)$, $y = y(t)$, we can adjust the chain rule to say:

$$\begin{aligned} df &= \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy \\ &= \frac{\partial f}{\partial x} \frac{dx}{dt} dt + \frac{\partial f}{\partial y} \frac{dy}{dt} dt \\ \implies \frac{df}{dt} &= \frac{\partial f}{\partial x} \frac{dx}{dt} + \frac{\partial f}{\partial y} \frac{dy}{dt} \end{aligned}$$

Example

$$f(x, y) = x^2 + y^2, \quad x(t) = t^2, \quad y(t) = t^3$$

Hence:

$$f(t) = t^4 + t^5 \implies \frac{df}{dt} = 4t^3 + 5t^4$$

By rewriting in terms of one variable:

$$\frac{\partial f}{\partial x} = 2x = 2t^2 \quad \frac{\partial f}{\partial y} = 2y = 2t^3$$

$$\frac{dx}{dt} = 2t \quad \frac{dy}{dt} = 3t^2$$

And instead using the new chain rule:

$$\begin{aligned} & \frac{\partial f}{\partial x} \frac{dx}{dt} + \frac{\partial f}{\partial y} \frac{dy}{dt} \\ &= (2t^2 + 2t) + (2t^3)(3t^2) \\ &= 4t^3 + 6t^5 \end{aligned}$$

Hence the new chain rule works!

2.1 Polar Coordinates

Suppose our x and y are now functions of two different variables themselves, so:

$$f = f(x, y) \quad x = x(r, \theta) \quad y = y(r, \theta)$$

From r, θ we want to calculate x, y and then from x, y we want to calculate f .

$$df = \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy$$

$$dx = \frac{\partial x}{\partial r} dr + \frac{\partial x}{\partial \theta} d\theta$$

$$dy = \frac{\partial y}{\partial r} dr + \frac{\partial y}{\partial \theta} d\theta$$

Hence:

$$df = \frac{\partial f}{\partial x} \left(\frac{\partial x}{\partial r} dr + \frac{\partial x}{\partial \theta} d\theta \right) + \frac{\partial f}{\partial y} \left(\frac{\partial y}{\partial r} dr + \frac{\partial y}{\partial \theta} d\theta \right)$$

$$df = \left(\frac{\partial f}{\partial x} \frac{\partial x}{\partial r} + \frac{\partial f}{\partial y} \frac{\partial y}{\partial r} \right) dr + \left(\frac{\partial f}{\partial x} \frac{\partial x}{\partial \theta} + \frac{\partial f}{\partial y} \frac{\partial y}{\partial \theta} \right) d\theta$$

We also know (if we substitute x, y into the original function to get a function of r, θ):

$$df = \frac{\partial f}{\partial r} dr + \frac{\partial f}{\partial \theta} d\theta$$

We can read off the final partial derivatives:

$$\frac{\partial f}{\partial r} = \frac{\partial f}{\partial x} \frac{\partial x}{\partial r} + \frac{\partial f}{\partial y} \frac{\partial y}{\partial r}, \quad \frac{\partial f}{\partial \theta} = \frac{\partial f}{\partial x} \frac{\partial x}{\partial \theta} + \frac{\partial f}{\partial y} \frac{\partial y}{\partial \theta}$$

As expected!

2.2 Generalising

Suppose we have two functions which map $\mathbb{R}^m \rightarrow \mathbb{R}^p$, and $\mathbb{R}^p \rightarrow \mathbb{R}^n$, respectively:

$$(x_1, x_2, \dots, x_m) \rightarrow (y_1, y_2, \dots, y_p) \rightarrow (z_1, z_2, \dots, z_n)$$

Then we have:

$$dz_i = \sum_{k=1}^p \frac{\partial z_i}{\partial y_k} dy_k$$

$$dy_k = \sum_{l=1}^m \frac{\partial y_k}{\partial x_l} dx_l$$

And substituting:

$$dz_i = \sum_{k=1}^p \sum_{l=1}^m \frac{\partial z_i}{\partial y_k} \frac{\partial y_k}{\partial x_l} dx_l$$

And (as the two sums are independent), we can pull out the inner sum:

$$= \sum_{l=1}^m \left[\sum_{k=1}^p \frac{\partial z_i}{\partial y_k} \frac{\partial y_k}{\partial x_l} \right] dx_l$$

$$= \sum_{l=1}^m \frac{\partial z_i}{\partial x_l} dx_l$$

The partial derivatives $\partial z_i / \partial x_j$ are therefore given by:

$$\frac{\partial z_i}{\partial x_j} = \sum_{k=1}^p \frac{\partial z_i}{\partial y_k} \frac{\partial y_k}{\partial x_j}$$

Fri 23 Jan 2026 12:00

Lecture 3 - Partial Differentiation III: Directional Derivatives and Gradients

1 Directional Derivatives and Gradients

Consider a scalar field $f(\underline{r}) = f(x, y, z)$. We can define derivatives along the x, y, z planes respectively:

$$\frac{\partial f}{\partial x} \quad \frac{\partial f}{\partial y} \quad \frac{\partial f}{\partial z}$$

Here, x, y, z aren't inherently special, they just form the standard basis we generally take. What if we want to compute the derivative at a point $\underline{r}$ with respect to a generic unit vector, $\hat{n}$, i.e. $\frac{\partial f}{\partial \hat{n}}$?

At our point $\underline{r}$, we are distance ϵ away from the direction formed by the $\hat{n}$ vector.

$$\frac{\partial f}{\partial \hat{n}} = \lim_{\epsilon \rightarrow 0} \frac{f(\underline{r} + \epsilon \hat{n}) - f(\underline{r})}{\epsilon}$$

This is a general form, called the directional derivative, giving the rate of change of the function in the direction $\hat{n}$. If we take the unit vector $\hat{n}$ to be $\hat{x}, \hat{y}, \hat{z}$ etc, this gives the standard derivatives in these directions.

If we know the derivatives in the x, y, z directions (or in any three directions which form an orthonormal basis), we can calculate the gradient of the function in any arbitrary direction.

$$\begin{aligned} \frac{\partial f}{\partial \hat{n}} &= \lim_{\epsilon \rightarrow 0} \frac{f(x + \epsilon n_x, y + \epsilon n_y, z + \epsilon n_z) - f(x, y, z)}{\epsilon} \\ &= \left. \frac{df}{d\epsilon}(x + \epsilon n_x, y + \epsilon n_y, z + \epsilon n_z) \right|_{\epsilon=0} \end{aligned}$$

This is just the chain rule:

$$\begin{aligned} &= \frac{\partial f}{\partial x} \frac{d}{d\epsilon}(x + \epsilon n_x) + \frac{\partial f}{\partial y} \frac{d}{d\epsilon}(y + \epsilon n_y) + \frac{\partial f}{\partial z} \frac{d}{d\epsilon}(z + \epsilon n_z) \\ &= n_x \frac{\partial f}{\partial x} + n_y \frac{\partial f}{\partial y} + n_z \frac{\partial f}{\partial z} \end{aligned}$$

Suppose we set $\hat{n}$ along x:

$$\begin{aligned} \hat{n} = \hat{i} &\implies n_x = 1, n_y = 0, n_z = 0 \\ \frac{\partial f}{\partial \hat{n}} &= \frac{\partial f}{\partial x} \end{aligned}$$

As expected!

It might be noticed that the expression is just a scalar product:

$$n_x \frac{\partial f}{\partial x} + n_y \frac{\partial f}{\partial y} + n_z \frac{\partial f}{\partial z} = \hat{n} \cdot \underline{\nabla} f$$

1.1 The Gradient

This ∇ “nabla” is called the gradient operator, it is the vector whose components are the partial derivatives in each direction:

$$\underline{\nabla} f \equiv \frac{\partial f}{\partial x} \hat{x} + \frac{\partial f}{\partial y} \hat{y} + \frac{\partial f}{\partial z} \hat{z} \equiv \underline{\text{grad}}(f)$$

We can also think of it as a general operator:

$$\underline{\nabla} \equiv \frac{\partial}{\partial x} \hat{x} + \frac{\partial}{\partial y} \hat{y} + \frac{\partial}{\partial z} \hat{z}$$

The directional derivative $\partial f / \partial n$ has the maximum magnitude when $\hat{n} \parallel \underline{\nabla} f$. The direction of $\underline{\nabla} f$ is the direction in which f increases most rapidly, and the magnitude is this maximum rate of increase.

1.2 Example

Suppose we have a particle moving in one dimension. Newton’s second law says:

$$m \frac{d^2 x}{dt^2} = F$$

Assuming that the motion arises from a conservative force, the force is defined as:

$$F = -\frac{dV}{dx}$$

Substituting this into Newton’s law, we obtain the equation of motion:

$$m \frac{d^2 x}{dt^2} + \frac{dV}{dx} = 0$$

We want to show that energy is conserved here. We know that total energy is:

$$E = \frac{1}{2} m v^2 + V(x(t)) = \frac{1}{2} m \left(\frac{dx}{dt} \right)^2 + V(x(t))$$

To show that total energy is constant, differentiate with respect to t :

$$\frac{dE}{dt} = \frac{d}{dt} \left[\frac{1}{2} m \left(\frac{dx}{dt} \right)^2 + V(x(t)) \right]$$

Using the chain rule:

$$\frac{dE}{dt} = m \frac{dx}{dt} \frac{d^2 x}{dt^2} + \frac{dV}{dx} \frac{dx}{dt}$$

Factoring out the velocity:

$$\frac{dE}{dt} = \frac{dx}{dt} \left[m \frac{d^2 x}{dt^2} + \frac{dV}{dx} \right]$$

Using the equation of motion, the bracketed term is zero:

$$\frac{dE}{dt} = \frac{dx}{dt} [0] = 0$$

Therefore, E does not change wrt time so is conserved.

1.3 Example in 3D

In 3D, we consider the components of force in each direction, again a conservative force (i.e. the electric field):

$$m \frac{d^2x}{dt^2} = F_x = -\frac{\partial V}{\partial x} \implies F_x + V_x = 0$$

$$m \frac{d^2y}{dt^2} = F_y = -\frac{\partial V}{\partial y} \implies F_y + V_y = 0$$

$$m \frac{d^2z}{dt^2} = F_z = -\frac{\partial V}{\partial z} \implies F_z + V_z = 0$$

Here, total energy is given by:

$$E(t) = \frac{1}{2}m \left[\left(\frac{dx}{dt} \right)^2 + \left(\frac{dy}{dt} \right)^2 + \left(\frac{dz}{dt} \right)^2 \right] + V(x(t), y(t), z(t))$$

$$\frac{dE}{dt} = m \frac{dx}{dt} \frac{d^2x}{dt^2} + m \frac{dy}{dt} \frac{d^2y}{dt^2} + m \frac{dz}{dt} \frac{d^2z}{dt^2} + \frac{\partial V}{\partial x} \frac{dx}{dt} + \frac{\partial V}{\partial y} \frac{dy}{dt} + \frac{\partial V}{\partial z} \frac{dz}{dt}$$

This gives:

$$\begin{aligned} \frac{dx}{dt} \left[m \frac{d^2x}{dt^2} + \frac{\partial V}{\partial x} \right] + \frac{dy}{dt} \left[m \frac{d^2y}{dt^2} + \frac{\partial V}{\partial y} \right] + \frac{dz}{dt} \left[m \frac{d^2z}{dt^2} + \frac{\partial V}{\partial z} \right] &= 0 \\ \frac{\partial x}{\partial t} [F_x + V_x] + \frac{\partial y}{\partial t} [F_y + V_y] + \frac{\partial z}{\partial t} [F_z + V_z] &= 0 \end{aligned}$$

This is zero as each of the bracketed terms is zero by N2L. We've done this laboriously component wise, but we can use the gradient to do it all at once:

$$m \frac{d\mathbf{r}}{dt} \cdot \frac{d^2\mathbf{r}}{dt^2} + \nabla V \cdot \frac{d\mathbf{r}}{dt}$$

Etc.

1.4 Numerical Example

Given $f(x, y, z) = x^3y^2z$. Find the rate of increase at $(1, 1, 1)$ in the direction given by $\hat{i} + 2\hat{j} + 2\hat{k}$.

$$\nabla f = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right) = (3x^2y^2z, 2x^3yz, x^3y^2) = (3, 2, 1) \text{ at } \underline{r} = (1, 1, 1)$$

And the direction is:

$$\mathbf{n} = (1, 2, 2)$$

However this does not have unit length, hence:

$$\hat{\mathbf{n}} = \frac{1}{\sqrt{1^2 + 2^2 + 2^2}}(1, 2, 2) = \frac{1}{3}(1, 2, 2)$$

So:

$$\frac{\partial f}{\partial n} = \nabla f \cdot \hat{\mathbf{n}} = \frac{1}{3}(3, 2, 1) \cdot (1, 2, 2) = \frac{1}{3}(3 + 4 + 2) = 3$$

2 Gradient and Vector Fields

We can also use Nabla on a vector field $\underline{E} = E_x\hat{i} + E_y\hat{j} + E_z\hat{k}$.

We have two possible ways of doing this, using the scalar product or the vector product. $\nabla \cdot \underline{E}$ is called the divergence and $\nabla \times \underline{E}$ is called the curl.

2.1 Divergence

$$\begin{aligned}\underline{\nabla} \cdot \underline{E} &= \left(\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z} \right) \cdot (E_x \hat{i} + E_y \hat{j} + E_z \hat{k}) \\ &= \frac{\partial E_x}{\partial x} + \frac{\partial E_y}{\partial y} + \frac{\partial E_z}{\partial z}\end{aligned}$$

This does have a geometric interpretation. Consider a small volume and calculate the flux of the field through that volume. The divergence gives the flux per unit volume.

2.2 Curl

$$\begin{aligned}\underline{\nabla} \times \underline{E} &= \left(\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z} \right) \times (E_x \hat{i} + E_y \hat{j} + E_z \hat{k}) = \det \begin{bmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ E_x & E_y & E_z \end{bmatrix} \\ &= \left(\frac{\partial E_z}{\partial y} - \frac{\partial E_y}{\partial z} \right) \hat{i} + \left(\frac{\partial E_x}{\partial z} - \frac{\partial E_z}{\partial x} \right) \hat{j} + \left(\frac{\partial E_y}{\partial x} - \frac{\partial E_x}{\partial y} \right) \hat{k}\end{aligned}$$

In a static steady state, the curl of the electric field is zero.

The proper study of gradient, divergence and curl is a second year topic (vector analysis). It's needed in Maxwell's equations for electromagnetism.

Wed 28 Jan 2026 11:00

Lecture 4 - Partial Differentiation IV: Total Time Derivative and Multivariate Taylor Series

1 Total Time Derivative

Consider a fluid with a velocity field:

$$\underline{v}(x, y, z, t) = v_x(x, y, z, t)\hat{i} + v_y(x, y, z, t)\hat{j} + v_z(x, y, z, t)\hat{k}$$

We could follow a small piece of this fluid, and use Newton's Second Law, per unit volume. This gives mass per unit volume (i.e. density) times acceleration. Assuming a non-compressible fluid with constant density:

$$\underline{f} = \rho \frac{d\underline{v}}{dt}$$

Where $\underline{f}$ is force per unit volume. Beginning by considering acceleration in only the x direction, there are two sources of time dependence. One from the fluid element itself moving, and one from the velocity of the piece changing with time.

This gives the total acceleration as:

$$\begin{aligned} \frac{dv_x}{dt} &= \frac{\partial v_x}{\partial x} \frac{dx}{dt} + \frac{\partial v_x}{\partial y} \frac{dy}{dt} + \frac{\partial v_x}{\partial z} \frac{dz}{dt} + \frac{\partial v_x}{\partial t} \\ &= v_x \frac{\partial v_x}{\partial x} + v_y \frac{\partial v_x}{\partial y} + v_z \frac{\partial v_x}{\partial z} + \frac{\partial v_x}{\partial t} \end{aligned}$$

We can simplify this with nabla:

$$\frac{dv_x}{dt} = \left(v_x \frac{\partial}{\partial x} + v_y \frac{\partial}{\partial y} + v_z \frac{\partial}{\partial z} \right) v_x + \frac{\partial v_x}{\partial t} = (\underline{v} \cdot \nabla) v_x + \frac{\partial v_x}{\partial t}$$

The total time derivative or advective derivative is given by:

$$\frac{d\underline{u}}{dt} = (\underline{v} \cdot \nabla) \underline{u} + \frac{\partial \underline{u}}{\partial t}$$

Where the first term gives the contribution from the element of fluid moving from the fluid flow pattern and the second term gives the contribution from the pattern itself changing.

The first term is called the advective term. A fluid element will accelerate even in a constant velocity field, because the velocity changes as it moves through the field.

The advective term is non-linear in the velocity field, which makes fluid mechanics difficult to solve.

2 Multivariate Taylor Series

For a single variable, we want to expand a function $f(x)$ around a point $x = a$ in a series of powers of h , where $h = x - a$:

$$f(a+h) = c_0 + c_1 h + c_2 h^2 + \dots = \sum_{m=0}^{\infty} c_m h^m$$

We find c_m by differentiating m times wrt h and setting $h = 0$:

$$f^m(a) = m!c_m \implies c_m = \frac{f^m(a)}{m!}$$

Hence:

$$f(a+h) = f(a) + hf'(a) + \frac{h^2}{2!}f''(a) + \dots = \sum_{m=0}^{\infty} \frac{h^m}{m!}f^m(a)$$

Or in operator form:

$$f(a+h) = \exp\left[h\frac{\partial}{\partial x}\right]f(a)$$

Noting that the exponential arises due to the Taylor series for the exponential function:

$$e^x = 1 + x + \frac{x^2}{2!} + \dots = \sum_{m=0}^{\infty} \frac{x^m}{m!}$$

$$\begin{aligned} f(a+h) &= f(x)|_{x=a} \\ &+ h\frac{d}{dx}f(x)\Big|_{x=a} \\ &+ \frac{h^2}{2!}\frac{d^2}{dx^2}f(x)\Big|_{x=a} \\ &\vdots \\ &= \left[1 + h\frac{d}{dx} + \frac{h^2}{2!}\frac{d^2}{dx^2} + \frac{h^3}{3!}\frac{d^3}{dx^3} \dots\right]f(x)\Big|_{x=a} \\ &= \exp\left[h\frac{d}{dx}\right]f(x)\Big|_{x=a} \end{aligned}$$

2.1 Multiple Variables

We want to expand a function $f(x, y)$ around the point $(x, y) = (a, b)$ as a power series.

This will take the form:

$$f(a+h, b+k) = c_{00} + c_{10}h + c_{01}k + c_{20}h^2 + c_{11}hk + c_{02}k^2 + \dots \sum_{m=0}^{\infty} \sum_{n=0}^{\infty} c_{mn}h^m k^n$$

We complete the same process to find coefficients, to find c_{mn} we differentiate m times wrt k , and n times wrt h and set $h = k = 0$ to get:

$$\frac{\partial^{m+n}}{\partial x^m \partial y^n}(a, b) = m!n!c_{mn}$$

Hence:

$$f(a+h, b+k) = \sum_{m=0}^{\infty} \sum_{n=0}^{\infty} \frac{h^m k^n}{m!n!} \frac{\partial^{m+n} f}{\partial x^m \partial y^n}(a, b)$$

Again in operator form this gives:

$$\exp\left[h\frac{\partial}{\partial x}\right]\exp\left[k\frac{\partial}{\partial y}\right]f(x, y)\Big|_{x=a, y=b}$$

2.2 Generalising to Many Variables

Consider a function of p variables $(x_1, x_2, \dots, x_p)$:

$$f(a_1 + h_1, \dots, a_p + h_p) = \sum_{n_1=0}^{\infty} \sum_{n_2=0}^{\infty} \dots \sum_{n_p=0}^{\infty} c_{n_1 n_2 \dots n_p} h_1^{n_1} h_2^{n_2} \dots h_p^{n_p}$$

Again to find the coefficients we differentiate n_1 times wrt h_1 , n_2 times wrt h_2 etc. We set $h_1 = h_2 = \dots = h_p = 0$ to get:

$$n_1! n_2! \dots n_p! c_{n_1 n_2 \dots n_p} = \frac{\partial^{n_1+n_2+\dots+n_p} f}{\partial x_1^{n_1} \partial x_2^{n_2} \dots \partial x_p^{n_p}}(a_1, \dots, a_p)$$

Hence:

$$f(a_1 + h_1, \dots, a_p + h_p) = \sum_{n_1=0}^{\infty} \dots \sum_{n_p=0}^{\infty} \frac{h_1^{n_1} \dots h_p^{n_p}}{n_1! \dots n_p!} \frac{\partial^{n_1+\dots+n_p} f}{\partial x_1^{n_1} \dots \partial x_p^{n_p}}(a_1, \dots, a_p)$$

In compact vector notation, where $\underline{h} = (h_1, \dots, h_p)$, this is:

$$f(\underline{a} + \underline{h}) = \exp \left[\sum_{i=1}^p h_i \frac{\partial}{\partial x_i} \right] f(\underline{a})$$

$$f(\underline{a} + \underline{h}) = \exp [\underline{h} \cdot \underline{\nabla}] f(\underline{a})$$

Thu 29 Jan 2026 09:00

Lecture 5 - Partial Differentiation V: 3D Stationary Points

1 2D Recap

In 2D, a stationary point of a function $f(x)$ occurs at $x = a$ where $f(x)$ is neither increasing or decreasing such that $f'(a) = 0$.

Consider the Taylor series about $x = a$, where $f'(a) = 0$:

$$f(a + h) = f(a) + hf'(a) + \frac{h^2}{2!}f''(a) + \dots = f(a) + \frac{h^2}{2!}f''(a) + \dots$$

If $f''(a) > 0$, the function looks like $+h^2$ near the stationary point. This gives us a minima. If the second derivative is negative the function looks like $-h^2$, giving a maximum point.

If the second derivative is zero, we need to find the first non-zero derivative $f^n(a) \neq 0$ where $n > 2$.

- If $n = 2k + 1$ (odd), the function looks like $\pm h^{2k+1}$ near the stationary point. This results in a point of inflexion.
- If $n = 2k$ (even) and $f^{2k}(a) > 0$, the function looks like $+h^{2k}$ near the stationary point so that we have a minimum.
- If $n = 2k$ (even) and $f^{2k}(a) < 0$, the function looks like $-h^{2k}$ near the stationary point so that we have a maximum.

2 Stationary Points of Several Variables

2.1 Two Variables

Consider a function $f(x, y)$ at a point (a, b) . For a stationary point to occur here, the function must be flat in all directions at that point, so all the partial derivatives must be zero:

$$f_x(a, b) = f_y(a, b) = 0$$

To classify it, we again consider the Taylor series:

$$f(a + h, b + k) \approx f(a, b) + \frac{1}{2}(Ah^2 + 2Bhk + Ck^2)$$

Where $A = f_{xx}(a, b)$, $B = f_{xy}(a, b)$ and $C = f_{yy}(a, b)$. We can then complete the square:

$$Ah^2 + Bhk + Ck^2 = A\left(h + \frac{B}{A}k\right)^2 + \left(C - \frac{B^2}{A}\right)k^2$$

- For the stationary point to be a minimum, we need $A > 0$ and $C - B^2/A = AC - B^2 > 0$.
- For a maximum, we need $A < 0$ and $C - B^2/A = AC - B^2 < 0$.
- For a saddle point, with a minimum in one direction and a maximum in the other, we need A and $C - B^2/A$ to be oppositely signed. This corresponds to the case where $AC - B^2 < 0$.

Considering the second derivatives directly:

- For a minimum we need $f_{xx} > 0$ and the Hessian determinant must be positive:

$$\Delta = \det H = \det \begin{bmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{bmatrix} > 0$$

- For a maximum, we need $f_{xx} < 0$ and the Hessian determinant must be positive:

$$\Delta = \det H = \det \begin{bmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{bmatrix} > 0$$

- For a saddle point, we need the Hessian determinant to be negative:

$$\Delta = \det H = \det \begin{bmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{bmatrix} < 0$$

2.2 Generalising To 3D

For a function $f(x, y, z)$ to have a stationary point at (a, b, c) , all first derivatives must be zero. The Taylor expansion then has no linear term, and performing the Taylor Expansion gives:

$$f(a+h, b+k, c+l) - f(a, b, c) = \frac{1}{2} (Ah^2 + 2Bhk + 2Chl + Dk^2 + 2Ekl + Fl^2)$$

where $A = f_{xx}$, $B = f_{xy}$, $C = f_{xz}$, $D = f_{yy}$, $E = f_{yz}$, $F = f_{zz}$, all evaluated at (a, b, c) . We define the quadratic:

$$Q = Ah^2 + 2Bhk + 2Chl + Dk^2 + 2Ekl + Fl^2$$

as the extra factor of $1/2$ does not change the classification of the point. The stationary point is a minimum $Q > 0$ for all $(h, k, l) \neq (0, 0, 0)$, because a positive Q means that the function increases in all directions away from the point.

Completing the square in h (treating k and l as fixed):

$$Ah^2 + 2Bhk + 2Chl = A \left(h + \frac{B}{A}k + \frac{C}{A}l \right)^2 - A \left(\frac{B}{A}k + \frac{C}{A}l \right)^2$$

Expanding the subtracted term:

$$-A \left(\frac{B}{A}k + \frac{C}{A}l \right)^2 = -A \left(\frac{B^2}{A^2}k^2 + \frac{2BC}{A^2}kl + \frac{C^2}{A^2}l^2 \right) = -\frac{B^2}{A}k^2 - \frac{2BC}{A}kl - \frac{C^2}{A}l^2$$

Substituting back and collecting the k - and l -terms:

$$Q = A \left(h + \frac{B}{A}k + \frac{C}{A}l \right)^2 + \underbrace{\left(D - \frac{B^2}{A} \right)}_{A'} k^2 + 2 \underbrace{\left(E - \frac{BC}{A} \right)}_{B'} kl + \underbrace{\left(F - \frac{C^2}{A} \right)}_{C'} l^2$$

The second quadratic in A' , $2B'$, C' is just our previous 2D example, so that condition holds. For the quadratic, we require:

$$\begin{aligned} A &> 0 \\ D - \frac{B^2}{A} &> 0 \\ \left(D - \frac{B^2}{A} \right) \left(F - \frac{C^2}{A} \right) - \left(E - \frac{BC}{A} \right)^2 &> 0 \end{aligned}$$

Expanding the last condition we get:

$$A^2DF - ADC^2 - AB^2F + B^2C^2 - E^2A^2 - B^2C^2 + 2EABC > 0$$

$$A(ADF - DC^2 - B^2F - E^2A + 2EBC) > 0$$

$$(ADF - DC^2 - B^2F - E^2A + 2EBC) > 0 \quad \text{as } A > 0$$

$$A(DF - E^2) - B(BF - EC) + C(BE - CD) > 0$$

$$\implies \det \begin{bmatrix} A & B & C \\ B & D & E \\ C & E & F \end{bmatrix} > 0$$

Finally, our conditions for a minimum are:

-

$$f_{xx} > 0$$

-

-

$$\det \begin{bmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{bmatrix} > 0$$

-

$$\det \begin{bmatrix} f_{xx} & f_{xy} & f_{xz} \\ f_{yx} & f_{yy} & f_{yz} \\ f_{zx} & f_{zy} & f_{zz} \end{bmatrix} > 0$$

And for a maximum:

-

$$f_{xx} < 0$$

-

-

$$\det \begin{bmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{bmatrix} > 0$$

-

$$\det \begin{bmatrix} f_{xx} & f_{xy} & f_{xz} \\ f_{yx} & f_{yy} & f_{yz} \\ f_{zx} & f_{zy} & f_{zz} \end{bmatrix} < 0$$

More generally, for a minimum we need these matrices, called the leading minors of the Hessian, to be positive and for a maximum we need them to alternate in sign.

Fri 30 Jan 2026 12:00

Lecture 6 - Partial Differentiation VI: Lagrange Multipliers

Note: Lecture began with a continuation of 3D stationary points, which I've bundled under last lectures notes to not interrupt the derivation.

1 Lagrange Multipliers

Suppose we wanted to minimise a function $f(x, y) = x^2 + y^2$ subject to a constraint, i.e. saying $x + y = 1$ (and therefore excluding the $(0, 0)$ trivial case).

While we could do this by writing $y = 1 - x$, substituting and minimising solely in terms of x , there is a more sophisticated method that becomes useful in more complex cases where the constraint is not invertible.

We consider the function:

$$F(x, y, \lambda) = x^2 + y^2 + \lambda(x + y - 1)$$

And minimise with respect to all three variables. When this happens, $F(x, y, \lambda) = f(x, y)$ at the minimised point.

Generally, to find the extrema of $f(x_1, x_2, \dots, x_n)$ subject to p constraints $g_i(x_1, x_2, \dots, x_n) = 0$, where $p < n$, we introduce a new function of $n + p$ variables:

$$F(x_1, x_2, \dots, x_n, \lambda_1, \lambda_2, \dots, \lambda_p) = f(x_1, x_2, \dots, x_n) + \sum_{i=1}^p \lambda_i g_i(x_1, x_2, \dots, x_n)$$

We then extremise F wrt all $n + p$ variables. Extremising with respect to λ_i enforces the constraint:

$$F_{\lambda_i} = g_i(x_1, x_2, \dots, x_n) = 0$$

So the the lambda term ends up dropping out and F and f have the same numerical value at extrema.

Wed 04 Feb 2026 11:00

Lecture 7 - Partial Differentiation VII: Statistical Mechanics

Recap of the last (two) lecture(s):

- To classify the stationary points of a function with n variables, we form the Hessian matrix $H_{ij} = f_{x_i x_j}$.
- From the top-left corner of this matrix, we form 1x1, 2x2 etc determinants. These are called the leading minors.
 - If these are all positive, the point is a maximum.
 - If they alternate in sign, we have a minimum.
 - If anything else, we have a saddle point.
- To extremise $f(x_1, x_2, \dots, x_n)$ subject to p constraints $g_i(x_1, x_2, \dots, x_n) = 0$, we introduce a new function of $n + p$ variables:

$$F(x_1, x_2, \dots, x_n, \lambda_1, \lambda_2, \dots, \lambda_p) = f(x_1, x_2, \dots, x_n) + \sum_{i=1}^p \lambda_i g_i(x_1, x_2, \dots, x_n)$$

- We then extremise F wrt all $n + p$ variables. Extremising with respect to λ_i enforces the constraint:

$$F_{\lambda_i} = g_i(x_1, x_2, \dots, x_n) = 0$$

- So the the lambda term ends up dropping out and F and f have the same numerical value at extrema.

1 Statistical Mechanics

Consider a system containing N atoms. Each atom has p energy levels, $\epsilon_1, \epsilon_2, \dots, \epsilon_p$.

Suppose there are n_1 atoms in state ϵ_1 , n_2 in state ϵ_2 , ..., n_p in state ϵ_p .

This gives us two constraints:

- The total number of atoms in the system is constant:

$$\sum_{i=1}^p n_i = N$$

- The total energy E of the system is conserved:

$$\sum_{i=1}^p n_i \epsilon_i = E$$

The entropy of the system is defined as:

$$S = k_B \ln W(n_1, n_2, \dots, n_p)$$

Where $W(\dots)$ represents the number of ways to put n_1 atoms in state ϵ_1 etc. We want to maximise this function subject to the two constraints. We assume that the atoms are indistinguishable.

The W function is given by the multinomial:

$$W(n_1, n_2, \dots, n_p) = \frac{N!}{n_1! n_2! \dots n_p!}$$

As, if all the atoms were in different energy levels, there would be $N!$ ways of doing so. However, the n_1 atoms occupying state ϵ_1 can be rearranged freely, with no actual change to the system. We have therefore overcounted by a factor of $n_1!$, so divide this out. Et cetera for each energy level.

From this:

$$\ln W = \ln N! - \ln n_1! - \ln n_2! - \dots - \ln n_p!$$

And:

$$\begin{aligned} \ln n! &= \ln 1 + \ln 2 + \dots + \ln n \\ &\approx \int_1^n \ln x \, dx \\ &= [x \ln x - x]_1^n = n \ln n - n \end{aligned}$$

Disregarding the lower bound as we assume that $n \gg 1$. This is called Stirling's approximation. We therefore have:

$$\ln W = (N \ln N - N) - \sum_{i=1}^p (n_i \ln n_i - n_i)$$

We are therefore looking to maximise:

$$F(\{n_i\}; \lambda_1, \lambda_2) = k_B \ln W + \lambda_1 \left(N - \sum_{i=1}^p n_i \right) + \lambda_2 \left(E - \sum_{i=1}^p n_i \epsilon_i \right)$$

And extremising wrt n_i gives:

$$\begin{aligned} \frac{\partial F}{\partial n_i} &= -k_B \ln n_i - \lambda_1 - \lambda_2 \epsilon_i = 0 \\ n_i &= \exp \left(-\frac{\lambda_1 + \lambda_2 \epsilon_i}{k_B} \right) \end{aligned}$$

Or:

$$n_i = \frac{1}{z} \exp \left(-\frac{\lambda_2 \epsilon_i}{k_B} \right)$$

where $z = \exp(-\lambda_1/k_B)$. The λ_2 has a physical meaning where $\lambda_2 = 1/T$.

We could also divide n_i/N to get the probability of finding a given atom in a given state:

$$p_i = \frac{1}{Z} \exp \left(-\frac{\lambda_2 \epsilon_i}{k_B} \right) \quad \text{where: } Z = \exp(-\lambda_1/k_B)/N$$

$$\sum_i p_i = 1$$

The normalisation constant Z is known as the partition function. The formulae for p_i gives the Boltzmann distribution.

2 Statistical Entropy

We have:

$$\begin{aligned} \frac{S}{k_B} &= \ln W = \ln N! - \sum_i \ln n_i! \\ &\approx (N \ln N - N) - \sum_i (n_i \ln n_i - n_i) \\ &= (N \ln N - N) - \sum_i (n_i \ln n_i) + N \end{aligned}$$

$$\begin{aligned}
&= (N \ln N) - \sum_i (n_i \ln n_i) \\
&= \sum_i (n_i) \ln N - \sum_i (n_i \ln n_i) \\
&= \sum_i (n_i \ln N - n_i \ln n_i) \\
&= \sum_i \left(n_i \ln \frac{N}{n_i} \right) \\
&= - \sum_i \left(n_i \ln \frac{n_i}{N} \right)
\end{aligned}$$

And multiplying through by N/N , we get:

$$\begin{aligned}
&= - \sum_i \left(\frac{N n_i}{N} \ln \frac{n_i}{N} \right) \\
&= -N \sum_i \frac{n_i}{N} \ln \frac{n_i}{N} \\
\frac{S}{k_B} &= -N \sum_i p_i \ln p_i
\end{aligned}$$

Hence the entropy per atom in units of k_B is:

$$\boxed{\frac{S}{Nk_B} = - \sum_i p_i \ln p_i}$$

this is known as the statistical entropy of the probability distribution $\{p_i\}$.

For a continuous distribution:

$$\frac{S}{Nk_B} = - \int p(x) \ln p(x) dx$$

3 Getting an Actual Value

We have (again considering per atom entropy):

$$p_i = \frac{1}{Z} \exp\left(-\frac{\lambda_2 \epsilon_i}{k_B}\right) = \frac{1}{Z} \exp\left(-\frac{\epsilon_i}{k_B T}\right)$$

$$\begin{aligned}
S &= -k_B \sum_i p_i \ln p_i \\
&= -k_B \sum_i p_i \left[-\ln Z - \frac{\epsilon_i}{k_B T} \right] \\
&= k_B \sum_i p_i \ln Z + k_B \sum_i p_i \frac{\epsilon_i}{k_B T} \\
&= k_B \ln Z \sum_i p_i + \frac{1}{T} \sum_i p_i \epsilon_i
\end{aligned}$$

The second sum is just average energy, which is the internal energy U .

$$\begin{aligned}
S &= k_B \ln Z + \frac{U}{T} \\
\implies U - TS &= k_B T \ln Z = F
\end{aligned}$$

Where “F” is called the Helmholtz Free Energy and the equation linking F and the partition function Z is called the bridge equation.

Thu 05 Feb 2026 09:00

Lecture 8 - Partial Differentiation VIII: Differentiation Under the Integral Sign

1 Differentiation Under the Integral Sign

Suppose we want to calculate the definite integral of a function which does not have an elementary antiderivative:

- Item

Fri 13 Feb 2026 12:00

Lecture 12 - End of Partial Differentiation & Start of ODEs

Recap of lecture 11:

- The tangent plane to a surface $f(x, y, z) = 0$ at (x_0, y_0, z_0) is given by:

$$\left(\frac{\partial f}{\partial x}\right)_0 (x - x_0) + \left(\frac{\partial f}{\partial y}\right)_0 (y - y_0) + \left(\frac{\partial f}{\partial z}\right)_0 (z - z_0) = 0$$

Such that $\nabla f(x_0, y_0, z_0)$ is the normal vector to the plane

- The parametric representation of a curve $\underline{r}(t)$ has:
 - Unit tangent: $\underline{\hat{T}} = \frac{d\underline{r}}{dt} / \left| \frac{d\underline{r}}{dt} \right|$
 - Arc length $s(t)$: $\frac{ds}{dt} = \left| \frac{d\underline{r}}{dt} \right| \rightarrow \underline{\hat{T}} = \frac{d\underline{r}}{ds}$.
 - Unit normal and curvature:

1 Orthonormal Triads

We can create an *orthonormal triad* by introducing a new normal vector called the unit binormal, $\underline{\hat{B}} = \underline{\hat{T}} \times \underline{\hat{N}}$.

Since $\underline{\hat{N}} \times \underline{\hat{N}} = \underline{0}$, differentiating wrt s gives:

$$\underline{\hat{N}} \cdot \frac{d\underline{\hat{N}}}{ds} = 0$$

TODO

We have:

$$\begin{aligned} \frac{d\underline{\hat{T}}}{ds} &= \kappa \underline{\hat{N}} \\ \frac{d\underline{\hat{N}}}{ds} &= -\kappa \underline{\hat{T}} + \tau \underline{\hat{B}} \end{aligned}$$

Hence:

$$\begin{aligned} \frac{d\underline{\hat{B}}}{ds} &= \frac{d}{ds} (\underline{\hat{T}} \times \underline{\hat{N}}) \\ &= \frac{d\underline{\hat{T}}}{ds} \times \underline{\hat{N}} + \underline{\hat{T}} \times \frac{d\underline{\hat{N}}}{ds} \\ &= \kappa \underline{\hat{N}} \times \underline{\hat{N}} + \underline{\hat{T}} \times (-\kappa \underline{\hat{T}} + \tau \underline{\hat{B}}) \\ &= \tau \underline{\hat{T}} \times \underline{\hat{B}} = \tau \underline{\hat{T}} \times (\underline{\hat{T}} \times \underline{\hat{N}}) \\ &= \tau [(\underline{\hat{T}} \cdot \underline{\hat{N}}) \underline{\hat{T}} - (\underline{\hat{T}} \cdot \underline{\hat{T}}) \underline{\hat{N}}] \\ &= \tau \underline{\hat{N}} \end{aligned}$$

This gives the Frenet-Serret Formulae:

This concludes partial differentiation! :D

2 Ordinary Differential Equations

A differential equation is any equation that involves derivatives. We care, because most laws of physics manifest themselves in the form of differential equations. For example:

$$\text{Newton's Second Law:} \quad \underline{F} = m \frac{d^2 \underline{r}}{dt^2}$$

$$\text{3D Time-Independent Schrödinger Eqn:} \quad -\frac{\hbar}{2m} \left(\frac{\partial^2 \psi}{dx^2} + \frac{\partial^2 \psi}{dy^2} + \frac{\partial^2 \psi}{dz^2} \right) + V(x, y, z) \psi = E \psi$$

$$\begin{aligned} \text{3D Wave Eqn:} \quad \frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} + \frac{\partial^2 \phi}{\partial z^2} &= \frac{1}{c^2} \frac{\partial^2 \phi}{\partial t^2} \\ &= \nabla^2 \phi = \frac{1}{c^2} \frac{\partial^2 \phi}{\partial t^2} \end{aligned}$$

$$\text{Gauss' Law:} \quad \frac{\partial E_x}{\partial x} + \frac{\partial E_y}{\partial y} + \frac{\partial E_z}{\partial z} = \frac{\rho}{\epsilon_0}$$

$$\text{Navier-Stokes Eqn:} \quad \rho \left(\frac{\partial \underline{v}}{\partial t} + (\underline{v} \cdot \nabla) \underline{v} \right) = -\nabla p + \rho \underline{g} + \mu \nabla^2 \underline{v}$$

In this course, we will only solve DEs of a single variable, i.e. Ordinary Differential Equations (ODEs). We don't look at Partial DEs of multiple variables yet.

In order to think about solving these, we need to classify them. Most DEs aren't soluble in closed form with elementary functions and need to be solved numerically. Here, we only consider nice soluble functions, but this is a vast minority in reality. We want to identify classes of DEs we can reasonably solve with a method for each.

We can generally solve linear equations by breaking them into small chunks and solving them individually, for example.

3 Types of DEs

3.1 Partial vs. Ordinary

In the examples above, only the first was an ODE, and the rest PDEs. Ordinary Differential Equations (ODEs) involve only a single variable.

Consider a vector $\underline{r}(t) = (x(t), y(t), z(t))$. t is called the independent variable, with x, y, z being dependant variables. While we have 3 dependant variables, we only have one independent variable (so only one thing to differentiate wrt), so this would end up being ordinary.

PDEs involve equations of two or more variables and hence involve partial derivatives.

3.2 Order

The order of a DE is given by the order of the highest derivative involved, so Newton's 2nd Law is a second order DE, as the highest order derivative is a second derivative.

3.3 Degree

The degree of a DE is a less important measure than the others. It is given by the highest power of the highest order derivative. For example, Newton's 2nd is a first degree, while an equation containing a^3 would be third degree (and second order, as a is a second derivative).

Ideally, we want this to be 1 for ease of solving, and higher degrees are rare but they do exist. For example, from Lagrangian Mechanics we have:

$$\frac{1}{2m} \left[\left(\frac{\partial s}{\partial x} \right)^2 + \left(\frac{\partial s}{\partial y} \right)^2 + \left(\frac{\partial s}{\partial z} \right)^2 \right] + V(x, y, z) = \frac{ds}{dt}$$

3.4 Homogenous and Inhomogeneous

A homogenous DE is a DE that does not have any terms of only the independent variable(s), while an inhomogeneous DE does.

For example, Newton's 2nd is homogenous as there is no term that involves t alone. This would be inhomogeneous:

$$\frac{\partial^2 x}{\partial t^2} = t + x$$

While this would be homogenous:

$$\frac{\partial^2 x}{\partial t^2} = tx$$

As t is a coefficient and not a pure term in its own right.

3.5 Linear and Non-Linear

A DE is linear if the dependant variable(s) and all of its/their derivatives occur purely as linear functions. For example:

$$(1 - x^2) \frac{d^2 y}{dx^2} - 2x \frac{dy}{dx} + n(n + 1)y = 0$$

This is linear, as the dependant variable y never has a power greater than 1.

$$\frac{dy}{dx} + xy = 0$$

Is also linear, while this is not:

$$\frac{dy}{dx} + xy^2 = 0$$

This is also non-linear (as shown by the Taylor Expansion of sine):

$$\frac{d^2 \theta}{dt^2} = -\frac{g}{l} \sin \theta$$

3.6 Examples

$$(1) \quad \left(\frac{\partial u}{\partial x} \right)^2 + \left(\frac{\partial u}{\partial y} \right)^2 = u^2$$

Homogenous first-order second-degree non-linear PDE.

$$(2) \quad \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = x^2 + y^2 + z^2$$

Inhomogeneous second-order first-degree linear PDE.

$$(3) \quad \frac{\partial y}{\partial x} + y^2 = x$$

Inhomogeneous first-order second-degree non-linear ODE.

Wed 18 Feb 2026 11:00

Lecture 13 - ODEs II

Summary of Lecture 12:

- Frenet-Serret Equations for a curve $\underline{r}(s)$, where $\hat{T} = d\underline{r}/ds$.
- Classification of differential equations:
 - Partial vs Ordinary
 - Order
 - Degree
 - Homogeneous vs Inhomogeneous
 - Linear vs Non-Linear

1 Equations Soluble By Direct Integration

Given some:

$$\frac{dy}{dx} = f(x) \implies y(x) = \int^x f(x') dx' + c$$

We can generalise this to some repeated derivative:

$$\frac{d^ny}{dx^n} = f(x)$$

Instead of having one undetermined constant here, we now have n . As each round of integration picks up a factor of the integration subject, these constants will form an n th degree polynomial.

1.1 Example: Particle Falling

$$\begin{aligned} \frac{d^2z}{dt^2} &= -g \\ \implies \frac{dz}{dt} &= -gt + v_0 \\ z &= -\frac{1}{2}gt^2 + v_0t + z_0 \end{aligned}$$

Here we consider our unknown constants as boundary conditions, i.e. the initial velocity and initial height.

2 Separable Equations

These are equations in the form:

$$\frac{dy}{dx} = \frac{f(x)}{g(y)}$$

$$g(y)dy = f(x)dx \implies \int g(y)dy = \int f(x)dx + c$$

2.1 Example I: Falling Particle with Air Resistance

$$\begin{aligned}
 \frac{dv}{dt} &= -g - kv \\
 \int \frac{dv}{g + kv} &= - \int dt \\
 \Rightarrow \frac{1}{k} \ln(g + kv) &= -t + c \\
 \Rightarrow k + gv &= Ae^{-kt} \\
 \Rightarrow v(t) &= -\frac{g}{k} + \left(v_0 + \frac{g}{k}\right)e^{-kt}
 \end{aligned}$$

2.2 Example II

$$\begin{aligned}
 \frac{dy}{dx} - x^2y^2 &= x^2 \\
 \frac{dy}{dx} &= x^2 + x^2y^2 = x^2(y^2 + 1) \\
 \Rightarrow \frac{1}{y^2 + 1} dy &= x^2 dx \\
 \Rightarrow \arctan y &= \frac{1}{3}x^3 + c \\
 \Rightarrow y &= \tan\left(\frac{1}{3}x^3 + c\right)
 \end{aligned}$$

2.3 Example III

$$\begin{aligned}
 \frac{dy}{dx} &= -\frac{x}{y} \\
 ydy &= -x dx \\
 \frac{1}{2}y^2 &= -\frac{1}{2}x^2 + c \\
 y^2 + x^2 &= 2c
 \end{aligned}$$

This is the equation for a circle.

3 Exact Equations

Suppose a function $y(x)$ is implicitly defined such that $f(x, y) = c$. It follows that the total differential:

$$df = \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy = 0$$

As $f(x, y)$ is a constant for all values of x, y , the constant differentiates to zero.

We can set:

$$M(x, y) = \frac{\partial f}{\partial x} \quad N(x, y) = \frac{\partial f}{\partial y}$$

To try to solve this equation:

$$M(x, y)dx + N(x, y)dy = 0$$

If we can do this (i.e. if the function can be decomposed into gradient components M and N), the equation is called “exact”. For this to be true, we need:

$$\frac{\partial^2 f}{\partial y \partial x} = \frac{\partial M}{\partial y} = \frac{\partial N}{\partial x} = \frac{\partial^2 f}{\partial x \partial y}$$

If this relationship holds, we have an exact equation and can integrate M and N to get $f(x, y)$ to solve the equation in the form $f(x, y) = c$

3.1 Example

$$\frac{dy}{dx} = -\frac{2x+y}{x+2y}$$
$$(2x+y)dx + (x+2y)dy = 0$$

So:

$$M(x, y) = 2x + y \quad N(x, y) = x + 2y$$

And:

$$\frac{\partial M}{\partial y} = 1 \quad \frac{\partial N}{\partial x} = 1$$

So yes, it is an exact function. It follows that:

$$\frac{\partial f}{\partial x} = 2x + y \implies f(x, y) = x^2 + xy + c(y)$$

$$\frac{\partial f}{\partial y} = x + 2y \implies f(x, y) = xy + y^2 + d(x)$$

$$\implies f(x, y) = x^2 + xy + y^2 = c$$

Thu 19 Feb 2026 09:00

Lecture 14

Recap of last lecture:

- Soluble by direct integration:

$$\frac{dy}{dx} = f(x) \implies y(x) = \int f(x)dx + c$$

- Separable equations:

$$\frac{dy}{dx} = \frac{f(x)}{g(y)} \implies \int f(x)dx = \int g(y)dy$$

- Exact equations:

Given in the form (or re-arrangeable to the form):

$$M(x, y)dx + N(x, y)dy = 0$$

These are soluble if:

$$M(x, y) = \frac{df}{dx} \quad N(x, y) = \frac{df}{dy}$$

With $f(x, y) = c$.

The condition for an equation being exact is:

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

1 “Homogeneous” Equations

Here, the word homogenous has a different meaning to classification of differentiating equations. If we have some function that satisfied this:

$$g(\lambda x, \lambda y) = \lambda^p g(x, y)$$

We call it a homogeneous function of order p .

“Homogeneous” differential equations in this case are functions of the form:

$$\frac{dy}{dx} = f\left(\frac{y}{x}\right)$$

Here, we make the substitution $v(x) = y(x)/x$. We can write:

$$y(x) = v(x)x$$

And substitute back in. We use the product rule where:

$$\begin{aligned} \frac{dy}{dx} &= x \frac{dv}{dx} + v = f(v) \\ x \frac{dv}{dx} &= f(v) - v \end{aligned}$$

This is now separable, so:

$$\int \frac{dv}{f(v) - v} = \int \frac{dx}{x}$$

1.1 Example

$$2xydy - (x^2 + y^2)dx = 0$$

This is not exact, so we put it into a standard form and substitute:

$$2xydy = (x^2 + y^2)dx$$

$$\frac{dy}{dx} = \frac{x^2 + y^2}{2xy} = \frac{(y/x)^2 + 1}{2(y/x)}$$

Let $u(x) = y/x$:

Fri 20 Feb 2026 11:56

Lecture 15 - Differential Equations III

Summary of last lecture:

- “Homogeneous” Equations

$$\frac{dy}{dx} = f\left(\frac{x}{y}\right) \rightarrow x \frac{dv}{dx} = f(v) - v \text{ where } y(x) = xv(x)$$

- Linear Equations

$$\sum_{k=0}^n a_k(x) \frac{d^k y}{dx^k} = f(x) \rightarrow y(x) = \sum_{k=1}^n \alpha_k y_k(x) + y_{PI}(x)$$

The solution is a sum of the complementary function (the general solution of the homogenous equation) and the particular integral (one solution of the inhomogeneous equation).

- First Order Linear Equations

$$\frac{dy}{dx} + P(x)y = Q(x) \rightarrow \frac{d}{dx} [y(x)e^{\int P(x)dx}] = Q(x)e^{\int P(x)dx}$$

1 Examples

1.1 Example I

$$(1 - x^2) \frac{dy}{dx} - xy = 1$$

Rewriting in the standard form:

$$\frac{dy}{dx} - \frac{x}{1 - x^2} y = \frac{1}{1 - x^2}$$

Hence:

$$P(x) = \frac{-x}{1 - x^2}$$

$$I(x) = \exp\left(-\int \frac{x dx}{1 - x^2}\right) = \exp\left(\frac{1}{2} \ln(1 - x^2)\right) = \sqrt{1 - x^2}$$

Multiplying through by the integrating factor:

$$\sqrt{1 - x^2} \frac{dy}{dx} - \frac{x}{\sqrt{1 - x^2}} y = \frac{1}{\sqrt{1 - x^2}}$$

The L.H.S is now the derivative of a product:

$$\frac{d}{dx} (y\sqrt{1 - x^2}) = \frac{1}{\sqrt{1 - x^2}}$$

And integrating both sides:

$$y\sqrt{1 - x^2} = \arcsin x + c$$

$$\Rightarrow \boxed{y = \frac{c}{\sqrt{1 - x^2}} + \frac{\arcsin x}{\sqrt{1 - x^2}}}$$

1.2 Example II

2 Linear ODEs with Constant Coefficients

In the most general form, we have:

$$a_n \frac{d^n y}{dx^n} + a_{n-1} \frac{d^{n-1} y}{dx^{n-1}} + \cdots + a_1 \frac{dy}{dx} + a_0 y = f(x)$$

Firstly, we solve the homogenous equation:

$$a_n \frac{d^n y}{dx^n} + a_{n-1} \frac{d^{n-1} y}{dx^{n-1}} + \cdots + a_1 \frac{dy}{dx} + a_0 y = 0$$

Let $y(x) = e^{\lambda x}$:

$$e^{\lambda x} (a_n \lambda^n + a_{n-1} \lambda^{n-1} + \cdots + a_1 \lambda + a_0) = 0$$

This reduces to finding the zeroes of the corresponding n th order polynomial. If this polynomial has n distinct zeroes, then the complimentary function is given by:

$$y_{CF}(x) = \alpha_1 e^{\lambda_1 x} + \alpha_2 e^{\lambda_2 x} + \cdots + \alpha_n e^{\lambda_n x}$$

If these roots contain a repeated root, this will reduce the number of unique solutions by one. If we have any complex solutions, they will come in complex conjugate pairs:

$$a^{(\pm ib)x} = e^{ax} (\cos bx \pm i \sin bx)$$

This therefore has independent real solutions:

$$e^{ax} \cos bx$$

$$e^{ax} \sin bx$$

3 Equidimensional Equations

These have coefficients which do depend on x , but where the coefficients are functions of x such that the n th derivative has a coefficient of $a_n x^n$

The general form of a homogeneous equidimensional equation is:

$$a_n x^n \frac{d^n y}{dx^n} + a_{n-1} x^{n-1} \frac{d^{n-1} y}{dx^{n-1}} + \cdots + a_1 x \frac{dy}{dx} + a_0 y = f(x)$$

This has a solution in the general form $y(x) = x^\lambda$. When we differentiate k times with respect to x we lose k powers of x , as each differentiation decreases the power. We need to restore this therefore with a x^k prefactor.

4 Mass on a Spring (SHM)

A mass, m on a spring is displaced from its equilibrium position by some distance x . There is a restoring force given by $F = -kx$.

$$F = ma \implies F = m \frac{d^2}{dt^2}$$

$$m \frac{d^2 x}{dt^2} + kx = 0$$

$$\frac{d^2 x}{dt^2} + \omega^2 x = 0 \quad \text{where } \omega^2 = \frac{k}{m}$$

We have:

$$x(t) = e^{\lambda t}$$

$$(\lambda^2 + \omega^2)e^{\lambda t} = 0$$

$$\lambda^2 + \omega^2 = 0$$

Hence:

$$\lambda = e^{\pm i\omega t}$$

Thu 05 Mar 2026 09:00

Lecture 20 - Methods of Integration

The derivative of an elementary function is always an elementary function. The integral of an elementary function is not always elementary however.
As a result, differentiation is always purely algorithmic churn, integration is often tricky.

Methods of integration are often based on methods of differentiation, for example:

- Integration by inspection: just spotting the answer.
- Integration by parts: Inverse of the product rule of differentiation.
- Integration by substitution: Inverse of the chain rule of differentiation.

1 Integration By Inspection

This is the most difficult to codify and can be easy for some people and very easy for others.

This might involve:

- Trigonometric Identities, for example replacing an expression whose integral is unknown with an expression whose integral is, using a trig sub.